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FACE RECOGNITION FOR ATTENDANCE

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ABSTRACT

In this paper we introduce an attendance management system. This system is based on face detection and recognition algorithms, which detects the student's face when he comes to the class room and marks him as present in the attendance by identifying him. The architecture and algorithms used at every stage are mentioned in this paper. Different scenarios are considered to check the efficiency of various face recognition tools.

KEYWORDS: Machine learning, Attendance, technology Methodology.

I. INTRODUCTION

In this 20Th century of automation many scientific upgrades and inventions have taken place to make the work less difficult, increase the precision and to trouble-free our lives. The Attendance System is the future that has taken place in the field of automation replacing older attendance marking activity. The Attendance Systems are generally finger print based, card based and web application based. These systems are used in different organizations. Old method of attendance marking is very time taking and becomes complex when the number of people are more. Automating the Attendance System has impact over default method as it saves time and also used for security purposes. This also helps to prevent proxy.

II. LITERATURE SURVEY

In this authors have proposed a finger print based attendance system. A easy to carry fingerprint device has been created which can be circulated among the students to place their finger on the scanner during the lecture time without the instructor's interference. This system usually guarantees a best method for marking the attendance. The cons with this method is that passing of the device during the lecture time may divert the concentration of the students.

A number of works related to Radio Frequency Identification (RFID) based Attendance Systems exist in the contrast. In this authors have proposed RFID based attendance system in which a student carry a RFID tag ID card and they will have to place that on the scanner to update their attendance. RS232 is used to establish the connection to the system in turn to the computer and save the recorded attendance from the database. This system may give rise to the problem of proxy access. An unauthorized student may make use of intended ID card and enter into the college.

III. PROPOSED MODEL

The system architecture is as shown in Figure 1. The proposed face detection attendance management system is based on face recognition algorithm. When a student enters the class his image is captured by the camera at the entrance. Face part is considered and is pre-processed for further processing part. Usually not more than five persons can enter the classroom at a given time face detection algorithm has prominently less work. Face Recognition proves to be having a lot of pros than other systems as discussed . When the student's face is recognized it is feed back to post-processing. The algorithm is being discussed.

The stages in the proposed Face recognition Attendance Management System are as shown below. Technical details of implementation of each stage are discussed in the next section

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A. Image Capture Phase

The Camera is placed at a distance from the entrance door to capture the front images of the students.

Algorithm Pseudo Code of Proposed System

1. Capture the Student's Image
2. Apply CNN algorithm(Convolutional neural network)
3. Now extract the required region in Rectangular Box
4. Next convert to gray scale, and Resize to 100x100
5. If Updation of Database is going on then

Store in Database

else

Apply compare_Faces(),if present it returns the name and get's updated in the attendance register

end if

6. Post-processing

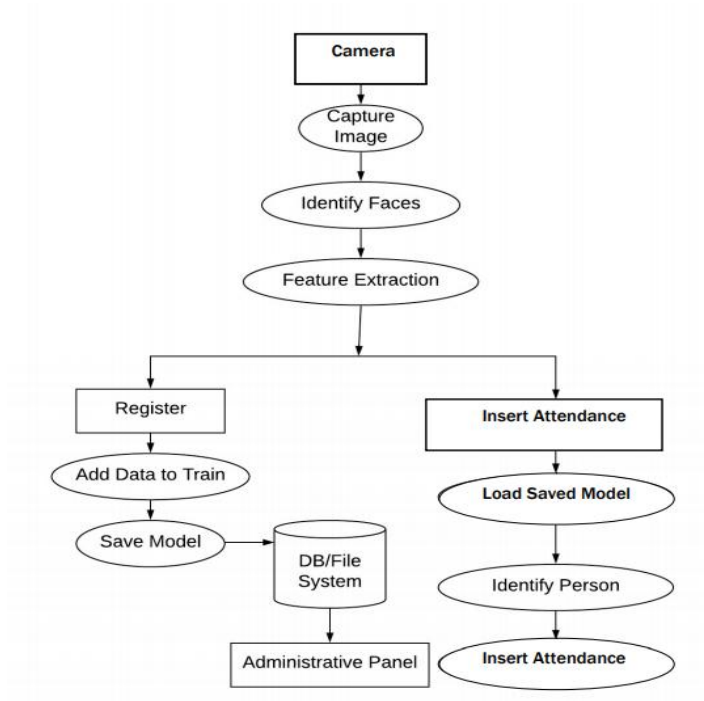

Fig-1: System Architecture

Fig-2: Feature extraction

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B. Face Detection Phase

A proper and efficient face detection algorithm such as CNN will be used to detect the face. As the algorithm is a model created by training a large set of database it's quite efficient and provides the accurate result. In feature in CNN uses the function named api_batch_face_locations() which creates a 2D array around the face to extract the features.

C. Pre-processing Phase

In the detected face using the 2D array the features are extracted and fed to pre- processing. The pre-processing step involves the comparing of face based on the encoding being provide by the 2D array which is usually 128 dimensions and the image is resized to 100*100.The function used for this process id compare_faces() and to get the encoding we use the function face_encoding().

D. Database Updating Phase

As we chose face recognition based system enrollment of every individual is required. This database updating phase consists the image of every individual and extracted features and later it is enhanced using pre-processing techniques and stored/updated in the database. In the project we have captured the images of individuals in different angles and also in different lighting conditions. If in case the lighting is too low the image cannot be captured so it's necessary for sufficient lighting conditions A database of 40 individuals with a single images of each has been collected for the working project.

E. Post-processing

In the proposed system, after recognizing the faces of the students, the names are updated into the database and can be displayed in the admin panel. The attendance is considered after 15 min after the class has begin. The system is also equipped with the facility of sending notification mail to the absentees when that facility is enabled. The admin has the power to send the mail to absentees. And an report is also available of the student attendance status just by clicking on the image.

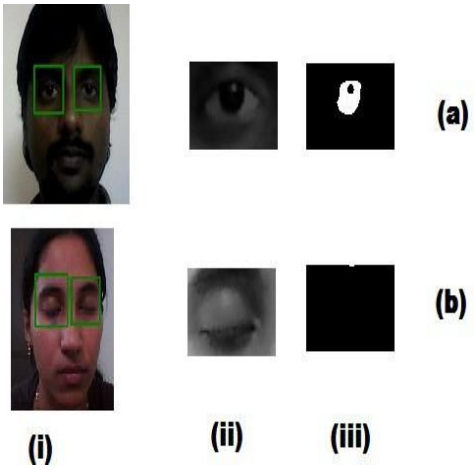

Fig-3: Iris and Eye region extraction

Fig-4: User Interface of the system proposed

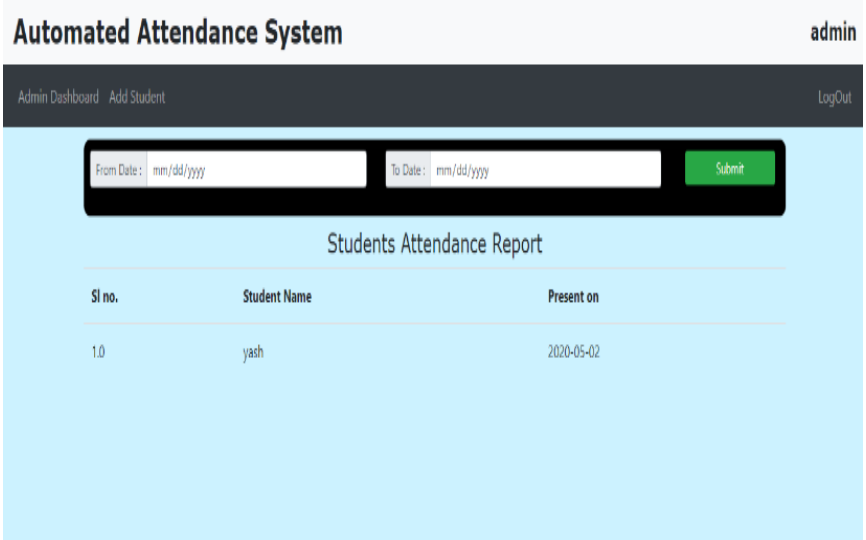

Fig-5: Student Attendance Report

IV. CONCLUSION

Face recognition Attendance Systems based on face recognition techniques using the machine learning models thus provide safe and secure attendance without any cons. This system can also be used to identify an unidentified people. In real time scenarios CNN is superior to other algorithms with better recognition efficiency and low false positive rate. The face_distance() acts as the classifier .

FUTURE WORK

The future work is to improve the recognition rate when there are some changes in a person like shaved head, using scarf, beard. CNN algorithm can be fused with face recognition systems in order to achieve better performance of the system.

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